Assignment 3

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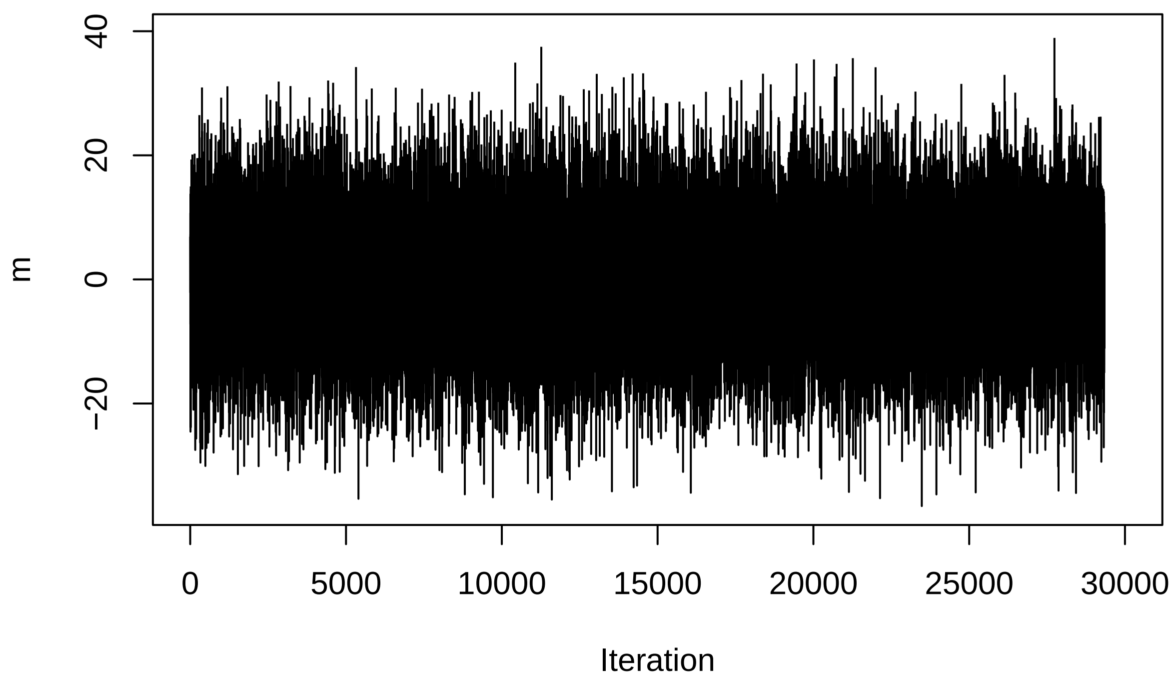
```r
###question 4

# Define the function to update theta
update_theta <- function(n, Y, Q, N, m) {
  theta <- rbeta(n, Y + exp(m) * Q, N - Y + exp(m) * (1 - Q))
  return(theta)
}

# Define the function to update m
update_m <- function() {
  m <- rnorm(1, 0, 10)
  return(m)
}

# MCMC algorithm
run_MCMC <- function(n, Y, Q, N, m_init, theta_init, iters) {
  # Chain initiation
  m <- m_init
  theta <- theta_init

  # Define chains
  m_chain <- rep(NA, iters)
  theta_chain <- matrix(NA, iters, n)

  # Start MCMC
  for (i in 1:iters) {
    m <- update_m()
    theta <- update_theta(n, Y, Q, N, m)
    m_chain[i] <- m
    theta_chain[i,] <- theta
  }

  # Return chains
  output <- list(m_chain = m_chain,
                 theta_chain = theta_chain)
  return(output)
}

# Data
n <- 10
Q <- c(0.845, 0.847, 0.880, 0.674, 0.909, 0.898, 0.770, 0.801, 0.802, 0.875)
Y <- c(64, 72, 55, 27, 75, 24, 28, 66, 40, 13)
```

```r
X <- c(75, 95, 63, 39, 83, 26, 41, 82, 54, 16)
N <- Y + X

# Run MCMC
MCMC_results <- run_MCMC(n = n, Y = Y, Q = Q, N = N,
                         m_init = 0,
                         theta_init = 1 - var(Y) / mean(Y),
                         iters = 30000)

# Plot chains
plot(MCMC_results$m_chain, xlab = "Iteration", ylab = "m", type = "l")
```

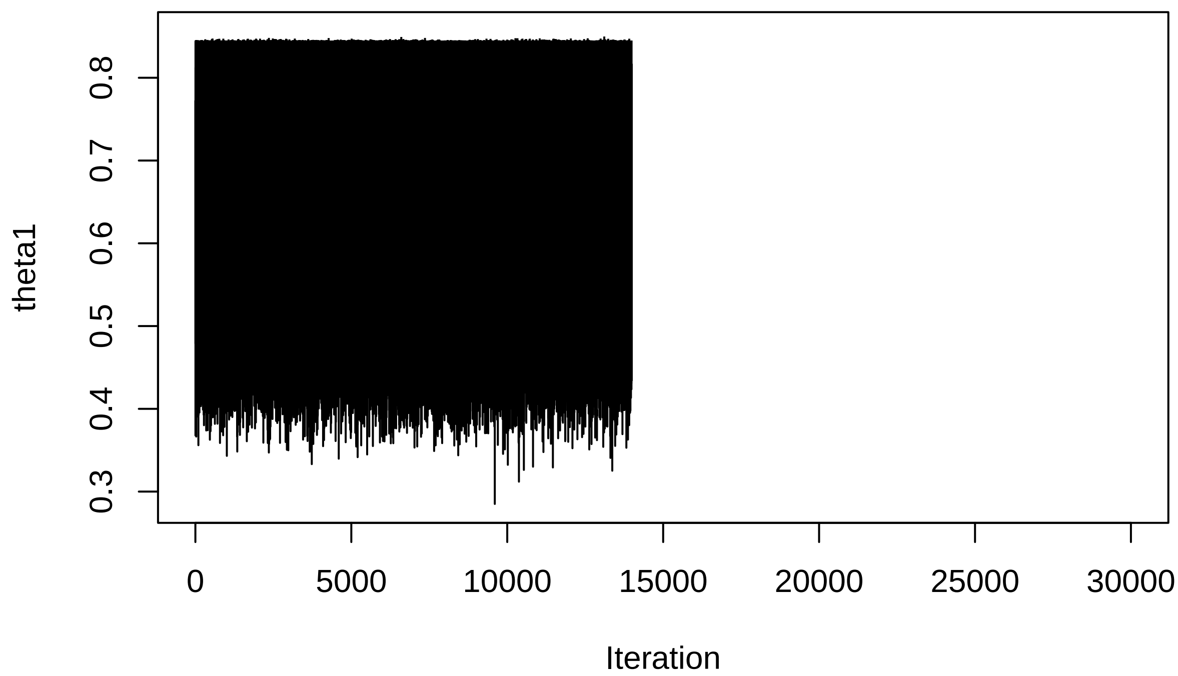



```r
plot(MCMC_results$theta_chain[, 1], xlab = "Iteration", ylab = "theta1", type = "l")

# Compute credible intervals
cred_intervals <- matrix(NA, 2, 11)
for (i in 1:10) {
  cred_intervals[, i] <- quantile(MCMC_results$theta_chain[, i], c(0.025, 0.975))
}
cred_intervals[, 11] <- quantile(MCMC_results$m_chain, c(0.025, 0.975))

# JAGS fitting
library(rjags)
```

```
## Warning: package 'rjags' was built under R version 4.2.3
```

```
## Loading required package: coda
```

```
## Warning: package 'coda' was built under R version 4.2.3

## Linked to JAGS 4.3.1

## Loaded modules: basemod,bugs
```

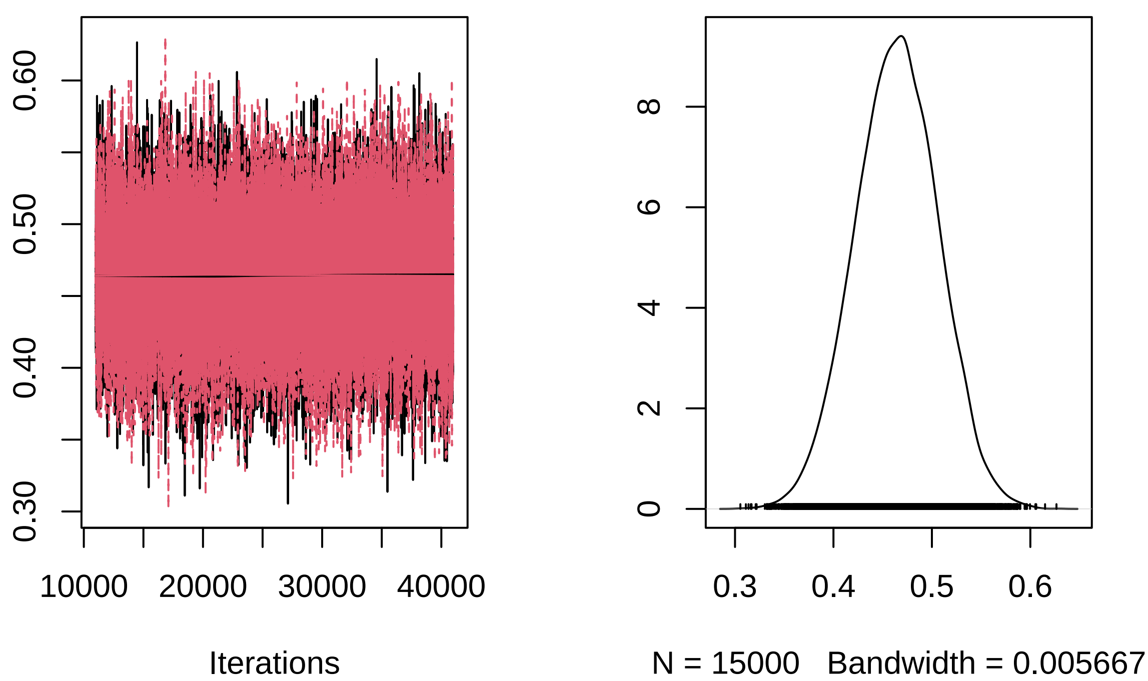



```r
data <- list(Y = Y, Q = Q, N = N, n = n)

model_string <- "
model{
  # Likelihood
  for(i in 1:n){
    Y[i] ~ dbin(theta[i], N[i])
  }

  # Priors
  for(i in 1:n){
    theta[i] ~ dbeta(exp(m) * Q[i], exp(m) * (1 - Q[i]))
  }
  m ~ dnorm(0, 10)
}"

model <- jags.model(textConnection(model_string), data = data,
                    n.chains = 2, quiet = TRUE)

update(model, 10000, progress.bar = "none")
```

```r
samples_theta <- coda.samples(model, variable.names = "theta",
                              n.iter = 30000, thin = 2, progress.bar = "none")
samples_m <- coda.samples(model, variable.names = "m",
                          n.iter = 30000, thin = 2, progress.bar = "none")

summary(samples_theta)[2]
```

```
## $quantiles
##                2.5%       25%       50%       75%     97.5%
## theta[1]  0.3814657 0.4358181 0.4645459 0.4929344 0.5449169
## theta[2]  0.3619860 0.4080970 0.4341511 0.4599109 0.5101369
## theta[3]  0.3824772 0.4404932 0.4707591 0.5016407 0.5603264
## theta[4]  0.3007214 0.3729584 0.4137735 0.4544283 0.5331449
## theta[5]  0.4015740 0.4514317 0.4779565 0.5045755 0.5554443
## theta[6]  0.3568099 0.4442465 0.4910640 0.5387235 0.6237334
## theta[7]  0.3006198 0.3732182 0.4122961 0.4524668 0.5284120
## theta[8]  0.3708602 0.4222186 0.4495450 0.4765726 0.5298241
## theta[9]  0.3346218 0.3966470 0.4308437 0.4652976 0.5318011
## theta[10] 0.2975074 0.4065296 0.4670478 0.5285816 0.6450390
```

```r
plot(samples_theta[, 1])
```




```r
summary(samples_m)[2]
```

```
## $quantiles
##       2.5%        25%        50%        75%      97.5%
## -0.2116798  0.1207438  0.2963991  0.4669042  0.7839768
```